

table

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1 Introduction

Table 1: Performance Comparison of Machine Learning Models for Fraud Detection

Model	Precision	Recall	F1 Score	Balanced Acc.	MCC	ROC-AUC	PR-AUC	Specificity	G-Mean
Random Forest	0.998181	0.998210	0.998190	0.838114	0.994041	0.996112	0.895284	0.999606	0.998907
XGBoost Classifier	0.998433	0.998589	0.998439	0.807852	0.995305	0.999937	0.930121	0.999668	0.999129
MLP	0.761200	0.627600	0.654000	0.627600	0.976300	0.978300	0.669800	0.997800	0.680400
Autoencoder + MLP	0.972000	0.973900	0.972300	0.497200	0.910900	0.996400	0.994000	0.900300	0.936400
Isolation Forest	0.499940	0.295110	0.371140	0.617563	0.294037	0.753493	0.369515	0.940016	0.526695
One-Class SVM	0.377528	0.712031	0.493432	0.736730	0.380729	0.814041	0.628104	0.761429	0.736316
Logistic Regression	0.951815	0.775140	0.841156	0.893757	0.597053	0.978701	0.978737	0.984761	0.873686
SGD Classifier	0.952766	0.876160	0.910261	0.686201	0.686201	0.966010	0.967408	0.986359	0.929628